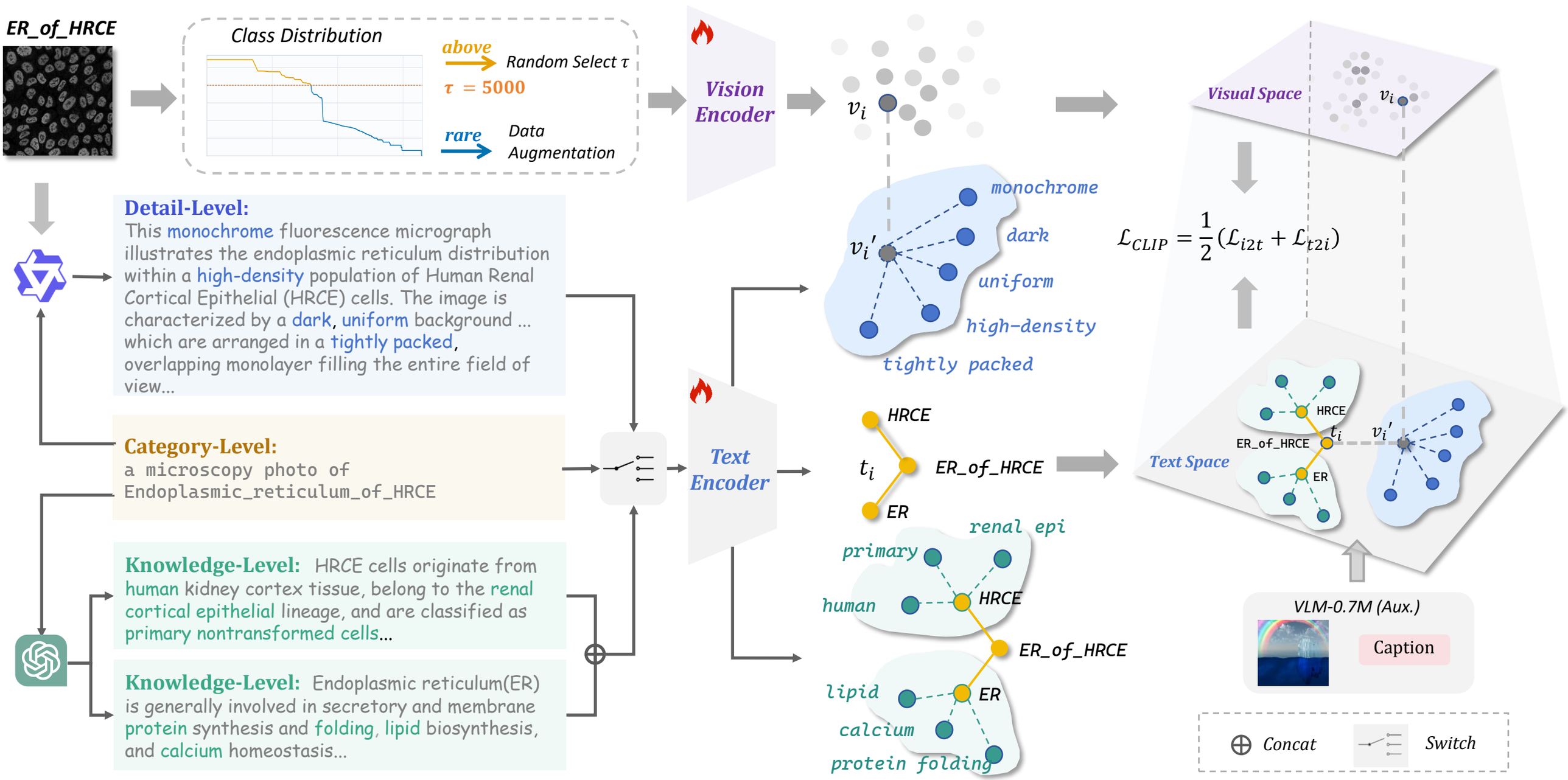

ER_of_HRCE
Class Distribution
above
Random Select τ
τ = 5000
rare
Data Augmentation
Vision Encoder
v_i
Visual Space
v_i
Detail-Level:
This monochrome fluorescence micrograph illustrates the endoplasmic reticulum distribution within a high-density population of Human Renal Cortical Epithelial (HRCE) cells. The image is characterized by a dark, uniform background ... which are arranged in a tightly packed, overlapping monolayer filling the entire field of view...
Category-Level:
a microscopy photo of Endoplasmic_reticulum_of_HRCE
Knowledge-Level: HRCE cells originate from human kidney cortex tissue, belong to the renal cortical epithelial lineage, and are classified as primary nontransformed cells...
Knowledge-Level: Endoplasmic reticulum(ER) is generally involved in secretory and membrane protein synthesis and folding, lipid biosynthesis, and calcium homeostasis...
Text Encoder
v_i'
monochrome
dark
uniform
high-density
tightly packed
t_i
HRCE
ER_of_HRCE
ER
renal epi
primary
human
HRCE
ER_of_HRCE
lipid
calcium
protein folding
ER
$\mathcal{L}_{CLIP} = \frac{1}{2}(\mathcal{L}_{i2t} + \mathcal{L}_{t2i})$
Text Space
HRCE
ER_of_HRCE
ER
t_i
v_i'
VLM-0.7M (Aux.)
Caption
Concat
Switch